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Plagiarism Detection: Identifying AI-Generated and Paraphrased Content

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INTRODUCTION

- The methods of detecting plagiarism have been refined in light of the increasing prevalence of AI-generated and paraphrased text.
- Today, plagiarism detection systems utilize techniques that rely on the semantics of text, its context, and its writing style.
- Such methods allow for ensuring the authenticity of material and maintaining integrity, while identifying whether any AI was used to generate or paraphrase the text.

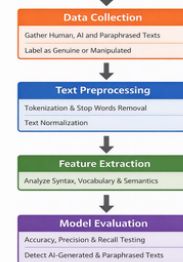
OBJECTIVES

- To create effective ways to detect the presence of AI-produced content as well as paraphrases based on linguistic, semantic, and structural elements in writing.
- To assess and compare different software tools currently available for identifying cases of plagiarism based on their efficiency, effectiveness, and capability of recognizing advanced rewriting methods utilized by AI system.
- To support academic integrity by creating efficient systems that can help educators differentiate between real human writing and AI-generated texts or paraphrases.

METHODOLOGY

- Data collection refers to the process of collecting a wide range of dataset with texts written by humans and created by AI along with paraphrased texts. The collection will involve labeling datasets as either genuine or manipulated. In this way, a basis will be formed to feed into the detection system.
- Text preprocessing entails processes like tokenizing and removing stop words and normalizing texts, ensuring that high-quality data is available for analysis.
- Feature extraction entails analyzing the dataset to detect patterns involving syntax structure, vocabulary use, syntactic structures, and semantic similarities using natural language processing and machine learning techniques.
- Model evaluation will involve testing the accuracy, precision, and recall of various models used to detect AI-created and paraphrased texts.

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ARCHITECTURE

- Dual classifiers are employed, whereby one classifier detects the patterns created by AI, and the second classifier determines the presence of paraphrases based on analysis of semantic shifts and rewriting.
- Feature fusion is done using a linguistic, semantic, and statistical combination of features that include syntax, vocabulary differences, embedding, and perplexity scores to enhance classification accuracy.

RESULT AND CONCLUSION

- Dual Classifier Approach : In this approach, there are two classifiers, where the first classifier detects artificial intelligence (AI) fingerprints, such as uniformity in structure, whereas the second classifier concentrates on paraphrasing the text using semantic changes. This allows the identification of texts written by humans, artificial intelligence, and paraphrases.
- Feature Fusion Module : The dual classifier approach includes linguistic, semantic, and statistical features, such as syntax, vocabulary richness, embeddings, and perplexity.

REFERENCES

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